IoT- ENABLED MULTI BIOMARKER CANCER RISK ASSESSMENT AND MONITORING SYSTEM

A PROJECT REPORT

Submitted by

ANCELIN MERBINA. M (962223205013)

JANITHA. J . R (962223205026)

STOJI HELBARIN . J (962223205058)

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Chunkankadai, Nagercoil - 629 003



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St. XAVIER'S CATHOLIC COLLEGE OF ENGINEERING

(An Autonomous Institution)
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BONAFIDE CERTIFICATE

Certified that this project report entitled **"IoT- ENABLED MULTI BIOMARKER CANCER RISK ASSESSMENT AND MONITORING SYSTEM "** is the bonafide work of **ANCELIN MERBINA. M (962223205013), JANITHA . J. R (962223205026), STOJI HELABARIN . J (962223205058)** who carried out the project work under my supervision.

SIGNATURE

Dr. Suja A. Alex B.Tech., M.E., Ph.D.
HEAD OF THE DEPARTMENT
Associate Professor
Department of Information Technology
St. Xavier's Catholic College of Engineering
Chunkankadai - 629 003

SIGNATURE

Dr. G Sahaya Stalin Jose M.E., Ph.D.
SUPERVISOR
Assistant Professor
Department of Information Technology
St. Xavier's Catholic College of Engineering
Chunkankadai - 629 003

Submitted for the viva-voce examination held at St. Xavier's Catholic College of Engineering on .....

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

Technology plays an important role in improving healthcare by making health assessment more accessible and efficient. This project aims to support cancer risk assessment and patient monitoring using two blood-based biomarkers: the Neutrophil-to-Lymphocyte Ratio (NLR) and the Monocyte-to-Lymphocyte Ratio (MLR). These ratios are associated with the body's inflammatory response and may provide useful information about the prognosis of certain cancers. The proposed system integrates image-processing techniques with embedded technology to support white blood cell analysis and simplify NLR and MLR assessment. Its cost-effective and portable approach aims to improve the convenience of monitoring biomarker variations, particularly in patients requiring regular medical follow-up.The system demonstrates the potential of technology-assisted healthcare monitoring. However, NLR and MLR values can also be influenced by infections and other health conditions. Therefore, the proposed system is intended to support cancer risk assessment and monitoring, not independently diagnose cancer or replace conventional laboratory investigations and professional medical evaluation.

CO-PO & PSO MAPPING

CO1: Identify any real world problem and find the optimal solution by applying gathered knowledge and skills.

CO	PO											PSO			SDGs	
	1	2	3	4	5	6	7	8	9	10	11	1	2	3	3	9
C01	3	3	3	2	3	2	2	3	2	2	2	3	3	3	3	3

3-High , 2-Medium , 1-Low

Mapping Justification

CO-PO Mapped	Justification
CO1 – PO1	Applies programming, mathematical calculations and computing knowledge to develop the proposed system.
CO1 – PO2	Identifies the challenges in obtaining accurate white blood cell counts and analyses possible ways to address them
CO1 – PO3	Develops a prototype by combining image processing and embedded hardware for NLR and MLR calculation.
CO1 – PO4	Tests the individual modules and examines the results to evaluate the system's performance.
CO1 – PO5	Uses Python, OpenCV, OV7670 camera, Arduino Uno and OLED display for system development.
CO1 – PO6	Addresses a healthcare-related problem by exploring technology-assisted blood biomarker assessment.
CO1 – PO7	Follows ethical practices during sample handling, data processing and presentation of project results.

CO1 – PO8	Encourages teamwork through the combined efforts of team members in software and hardware development.
CO1 – PO9	Communicates project details through technical documentation, diagrams and presentations.
CO1 – PO10	Involves planning, task allocation and management of resources during project development.
CO1 – PO11	Encourages learning about image processing, embedded systems and new technologies related to healthcare applications.

CO-PSO MAPPED	JUSTIFICATION
CO1 – PSO1	Applies programming skills using Python and embedded programming to develop the proposed system.
CO1 – PS02	Uses IT infrastructure, image-processing tools and analytical techniques to address a healthcare-related problem.
CO1 – PS03	Integrates camera-based image acquisition, software processing, Arduino and OLED display to develop a computing prototype.

SDG	JUSTIFICATION
Good Health and Well-being	Supports cancer-related health assessment research through the calculation of NLR and MLR values for monitoring purposes.
Industry, Innovation and Infrastructure	Combines image processing, camera technology and embedded hardware to develop a healthcare-related prototype.

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LIST OF SYMBOLS ABBREVIATIONS

S. No.	Abbreviation / Symbol	Description
1	IoT	Internet of Things
2	NLR	Neutrophil-to-Lymphocyte Ratio
3	MLR	Monocyte-to-Lymphocyte Ratio
4	WBC	White Blood Cell
5	CMOS	Complementary Metal-Oxide-Semiconductor
6	LED	Light-Emitting Diode
7	USB	Universal Serial Bus
8	IDE	Integrated Development Environment
9	OpenCV	Open Source Computer Vision Library
10	OLED	Organic Light-Emitting Diode
11	ADC	Analog-to-Digital Converter
12	Python	Programming Language Used for Image Processing
13	Arduino Uno	Microcontroller Development Board
14	OV7670	CMOS Camera Module
15	SH1106G	OLED Display Controller
16	N	Neutrophil Count
17	L	Lymphocyte Count
18	M	Monocyte Count
19	NLR	N/L
20	MLR	M/L

CHAPTER 1

INTRODUCTION

1.1 Introduction

Cancer is one of the major health concerns worldwide, making early assessment and continuous patient monitoring important aspects of healthcare. Blood-based biomarkers are widely studied to understand changes in the body's immune and inflammatory responses. Two such biomarkers, the Neutrophil-to-Lymphocyte Ratio (NLR) and Monocyte-to-Lymphocyte Ratio (MLR) , have been investigated for their relationship with cancer outcomes and treatment effectiveness.

NLR represents the ratio of neutrophils to lymphocytes, while MLR represents the ratio of monocytes to lymphocytes. Variations in these ratios may provide additional information about a patient's health condition when interpreted alongside other clinical findings. However, these values can also be influenced by infections, inflammation, medications and other medical conditions.

Conventional blood-cell analysis generally requires laboratory facilities and trained personnel. Advances in the Internet of Things (IoT), image processing and embedded technology offer opportunities to explore more convenient and cost-effective approaches to health monitoring.

This project aims to develop a prototype for calculating the Neutrophil-to-Lymphocyte Ratio (NLR) and Monocyte-to-Lymphocyte (MLR) . It uses blood sample images to obtain the required cell counts and calculate the ratios and display the results.

The proposed setup includes an OV7670 camera to capture the images , Python and OpenCV to process them and an Arduino Uno connected to an OLED display. The main purpose of this project is to study how technology can be used to calculate and display the NLR and MLR values.

However the proposed system is only intended to support the assessment of these blood parameters. It cannot diagnose cancer on its own or replace test performed in a medical laboratory.

1.2 Problem Statement

Traditional blood cell analysis is usually carried out in laboratories with the help of specialised equipment and trained staff . For patients who need regular checkups depending on laboratory tests may sometimes make frequent monitoring difficult.

NLR and MLR are two blood based ratios that researchers have studied to understand their possible relationship with cancer prognosis . Calculating these ratios requires accurate count of Neutrophils , Lymphocytes , and Monocytes . However identifying these different types of white blood cells from images can be challenging particularly when the images are unclear or have low resolution.

Therefore this project aims to explore how image processing and embedded technology can be used together to analyze blood- cell images and calculate NLR and MLR values . The proposed system is intended to offer a simple and affordable way to display the calculated results. Further testing and comparison with standard laboratory results are needed to determine the accuracy of the system.

1.3 Objectives of the project:

The main objective of this project are :

1. To develop a prototype that uses blood cell images to calculate NLR and MLR values.
2. To study how image processing techniques can be used to analyse white blood cells.
3. To obtain Neutrophil , Lymphocyte , and Monocyte counts required for the calculations.
4. To calculate the NLR and MLR based on the respective blood counts.
5. To connect the image processing system with an Arduino Uno and OLED display to present the results.
6. To examine the possibility of using IoT Technology for easier biomarker monitoring.
7. To identify the limitations of the proposed system and suggest improvements for future development.

1.4 Scope of the Project

This project focuses on developing a prototype for calculating NLR and MLR using blood cell images . It includes the preparation of the blood sample capturing the image , processing the image , transferring the cell count data , calculating the ratios and displaying the results.

The proposed system uses an OV7670 camera to capture the blood sample image.Python and OpenCV are used for image processing while Arduino Uno performs the calculations and control the OLED display to show the results.

The scope of the project is limited to the features developed and tested in the prototype. The accuracy of the NLR and MLR values depends on how

reliably the system identifies and counts the different types of white blood cells . Improving image quality identifying cells more accurately and comparing the results with standard laboratory measurements are important steps before considering any medical application.

The system is not designed to diagnose cancer independently. Instead the project explores how image processing and embedded technology could support the NLR and MLR assessment and contribute to the development of healthcare monitoring systems in the future.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter discusses the previous studies on the Neutrophil-to-Lymphocyte Ratio (NLR) and Monocyte-to-Lymphocyte Ratio (MLR) and the use of computer based techniques to analyse blood cell images . The reviewed studies investigate the possible role of inflammatory blood-cell ratios in cancer prognosis and the use of image-processing methods to support cell recognition. The studies are summarised using the research focus, technology or method, advantages, and limitations relevant to the proposed mini-project.

2.2 Review Of Existing Studies

2.2.1 NLR and MLR as Prognostic Biomarkers in Cancer

Research Paper : Ying -Ying Zhang et al., "Associations between Peripheral Whole Blood Cell Counts Derived Indexes and Cancer Prognosis: An Umbrella Review of Meta-Analyses of Cohort Studies," Critical Reviews in Oncology/Hematology, 2024.

Method Used: The authors reviewed previously published meta-analyses to examine the relationship between blood cell based indicators , including NLR and other related ratios and cancer prognosis.

Advantages:

- The review combines findings from several published meta-analyses , providing a broader understanding of the research .

- It explains why blood cell ratios are being studied as possible indicators of cancer prognosis.

Disadvantages:

- The findings different across studies making it difficult to draw consistent conclusions.
- The results may not apply equally to all types of cancer or patients.

2.2.2. Monocyte-to-Lymphocyte Ratio in Cancer Prognosis

Research Paper: Tanaka et al., "Monocyte-to-Lymphocyte Ratio Is a Prognostic Predictor for Patients with Non-Small Cell Lung Cancer Treated with Stereotactic Body Radiation Therapy," 2024.

Method Used: The researchers studied whether the Monocyte-to-Lymphocyte Ratio (MLR) was related to treatment outcomes in patients with non-small cell lung cancer who received stereotactic body radiation therapy.

Advantages:

- The study examines whether MLR can provide additional information about cancer prognosis.
- It focuses on patients with a specific type of lung cancer who received a particular treatment.

Disadvantages:

- The findings may not be applicable to patients with other types of cancer .
- MLR values may also change due to infections , inflammation and other health conditions which can affect their interpretation.

2.2.3. Image Processing and Machine Learning for Blood-Cell Analysis

Research Paper: José Rodellar et al., "Image Processing and Machine Learning in the Morphological Analysis of Blood Cells," International Journal of Laboratory Hematology, 2018; 40(Suppl. 1):46–53. DOI: 10.1111/ijlh.12818.

Method Used: The review explain how blood cell images are analyzed using image segmentation feature extraction and classification techniques. These methods help identify and examine the features of cells in peripheral blood smear images

Advantages:

- It provides a clear explanation of the main steps involved in automated blood cell images analysis.
- It show how computer based methods can help make the analysis of cell shapes and structure more efficient and consistent

Disadvantages:

- Automated blood cell identification still requires further testing to ensure accurate results.
- The methods discussed in the review may focus on specific types of abnormal cells and may not be directly suitable for the proposed project.

2.2.4. Blood-Cell Image Segmentation and Classification

Research Paper: Shahzad et al., "Blood Cell Image Segmentation and Classification: A Systematic Review," 2024

Method Used: The review examines different methods used to analyze images of red and white blood cells . It discusses techniques for separating cells from the background, identifying important cell features and classifying the cells . The review also covers methods that use machine learning and deep learning for blood cell analysis.

Advantages:

- The review summarises the commonly used methods for analysing blood cell images and explains the factors considered when evaluating these methods.
- It highlights the importance of cell morphology and using suitable image dataset for blood cell analysis.

Disadvantages:

- Different studies use different dataset and image acquisition methods , which can make comparison difficult.
- Overlapping cells images related errors and variations in cell shape can make accurate cell classification challenging.

2.2.5. Artificial Intelligence for Peripheral Blood-Film Analysis

Research Paper: "From Microscope to Micropixels: A Rapid Review of Artificial Intelligence for the Peripheral Blood Film," published in 2023

Method Used: The review discusses previous studies that used artificial intelligence to identify and classify cells in peripheral blood smear images.

Advantages:

- The review shows that automated blood film analysis is gaining interest in different areas of research and healthcare.
- It highlights the importance of testing and validating image based systems to ensure that they produce reliable results.

Disadvantages:

- The result reported in previous studies may not be the same when the methods are used with a different camera or dataset.
- For practical purposes the system needs standardised data and proper validation under different conditions.

2.3 Comparative Summary

Study / Area	Main Method	Contribution	Main Limitation
Blood-count indexes and cancer prognosis (2024)	Umbrella review of meta-analyses	Summarises evidence on blood-cell ratios and cancer outcomes	Evidence and effect directions vary
MLR in lung cancer (2024)	Clinical prognostic analysis	Investigates MLR in a defined treatment group	Limited generalisation to other groups
Rodellar et al. (2018)	Segmentation, feature extraction, classification	Explains computer-based cell morphology analysis	Further validation remains necessary
Shahzad et al. (2024)	Segmentation, feature extraction, classification	Compares methods and dataset practices	Data variability and image artefacts
Peripheral blood-film AI review (2023)	Rapid review of AI-based classification	Surveys reported models and evaluation	Real-world validation is still needed

2.4 Research Gap

The reviewed studies show that NLR and MLR are being examined for their possible role in predicting cancer outcomes. At the same time research on image processing shows that computers can be used to analyze blood cell images. However obtaining accurate cell counts remains a major challenge because image quality , camera focus , lighting , overlapping cells , and identification errors can affect the results. NLR and MLR can also vary due to conditions other than cancer which means these ratios alone cannot confirm the presence of cancer.

This project aims to develop a prototype that combines camera based image capture , image processing , ratio calculation and result display . The accuracy of the calculated values needs to be checked by comparing them with reliable cell coins and standard laboratory results. Therefore the proposed system is a technology based prototype and should not be considered a clinically validated tools for diagnosing cancer.

CHAPTER 3

PROPOSED SYSTEM

3.1 Overview

The proposed system focuses on using two blood based indicators the Neutrophil-to-Lymphocyte Ratio (NLR) and the Monocyte-to-Lymphocyte Ratio (MLR). The system uses a camera to capture blood sample images, processes the images and uses embedded hardware to calculate and display the NLR and MLR values.

The prepared blood sample is photographed using an OV7670 CMOS camera with the help of LED lighting. The captured images are processed on a computer using Python and the OpenCV library to obtain the relevant white blood cell count information. After image processing the resulting data is sent to the Arduino Uno through serial communication.

The Arduino Uno calculates NLR and MLR using the received cell counts and display the result on the OLED screen. The system combines a camera , image processing software and electronic components to calculate and show these ratios.

The accuracy of the results depends on image quality and correct cell counts. Further testing and comparison with laboratory results are needed before using the system for medical purposes.

3.2 System Architecture

The system architecture shows how the hardware and software components work together . It includes preparing the blood sample, capturing and processing the image, transferring the data through serial communication, calculating the ratios and displaying the results.

The blood sample is first prepared using the proper laboratory procedure and placed under the camera. The OV7670 camera captures the sample with LED lighting. The image is then sent to a computer where Python and OpenCV are used to process it .

The image is processed to obtain the required cell counts . These counts are sent to the Arduino Uno through serial communication. The Arduino then calculates the NLR and MLR values and displays the results on the OLED screen.

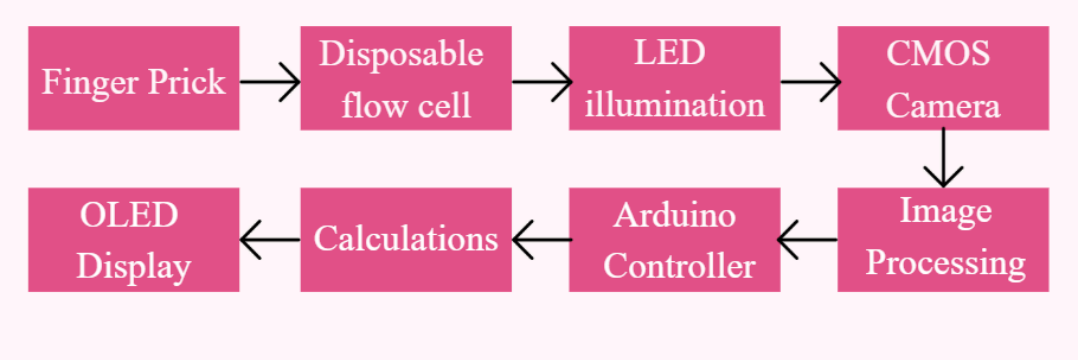


Figure 3.1: System Architecture of the Proposed System

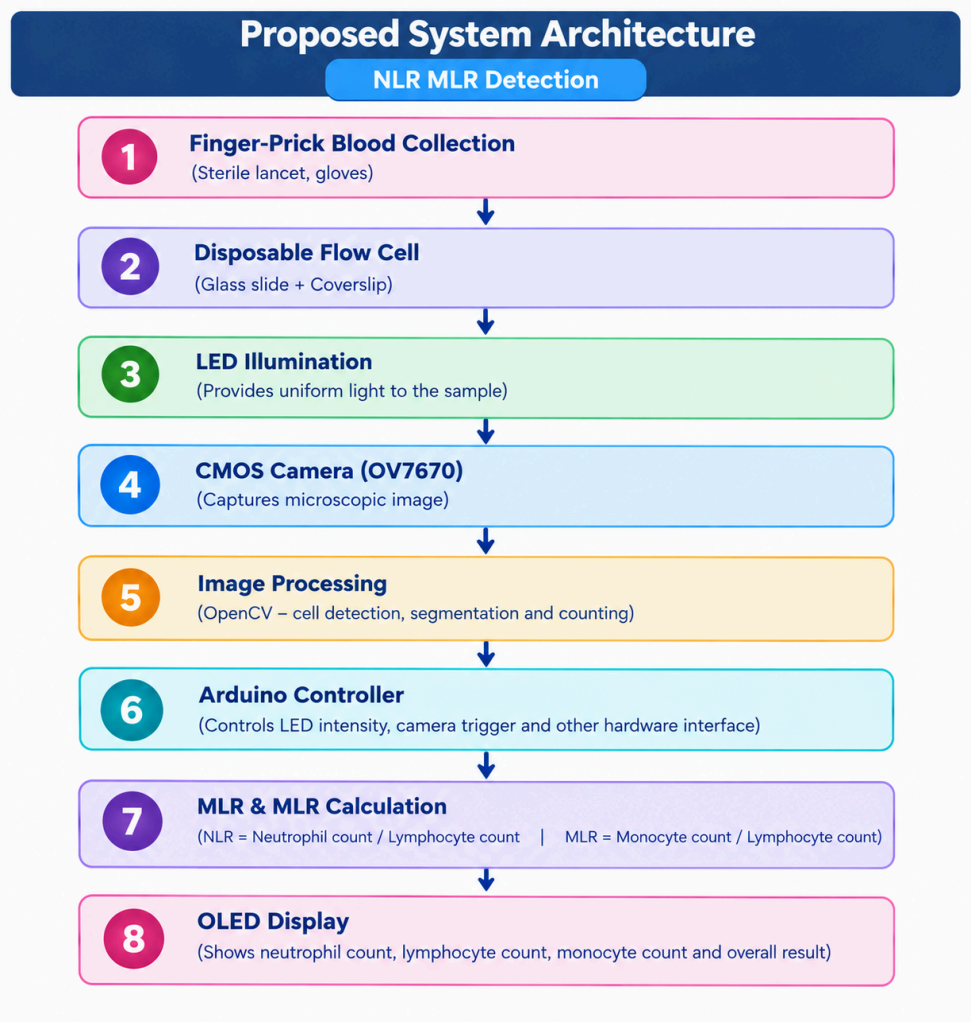
The system connects image analysis with the electronic components. Each component has its own task and works together to make the prototype function properly.

3.2.1 Block Diagram of the Proposed System

The block diagram shows how the system works , starting with blood sample preparation and ending with the display of the calculated ratios.

The camera captures an image of the prepared blood sample which is then processed on a computer using Python and OpenCV. The cell-count data is sent to the Arduino Uno , where the NLR and MLR values are calculated. Finally the results are shown on the OLED screen.

Figure 3.2: Block Diagram of the Proposed System



The main processing sequence is:

Blood Sample Preparation → OV7670 Camera and LED Illumination → Image Processing Using Python and OpenCV → Serial Communication → Arduino Uno → NLR and MLR Calculation → OLED Display

3.2.2 Flowchart of the Proposed System

The flowchart shows the steps involved in the proposed system , starting with blood sample preparation , followed by image capture and image processing. The cell count data are then sent to the Arduino Uno to calculate NLR and MLR . Before calculating the ratios, the program checks whether the received data are valid . If the data are invalid or the lymphocyte count is zero the system should not display the ratios. Otherwise the calculated values are shown on the OLED screen.

3.3 Module Description

The proposed system has separate modules for preparing the blood sample , capturing and processing images , transferring data , calculating NLR and MLR and displaying the results.

3.3.1 Blood Sample Preparation Module

The blood sample preparation module prepares the sample for image acquisition. Appropriate sample preparation is important because the visibility and appearance of blood cells affect the quality of the captured image.

The sample is prepared using the procedure defined for the prototype and positioned in the image acquisition arrangement. Where Turk's solution is used, its preparation and application must follow the approved laboratory procedure.

Preparing the blood sample in the same way each time helps maintain consistent image quality and makes image analysis easier. Proper care should also be taken when handling and disposing of blood samples to ensure safety.

3.3.2 Image Acquisition Module

The image acquisition module uses the OV7670 CMOS camera to capture the prepared blood sample . LED lighting helps illuminate the sample so that it can be viewed by the camera . The captured image is then transferred to a computer using the camera acquisition software for further processing.

The quality of the image depends on proper lighting camera position, focus and sample preparation . Clear images are important for image processing as poor lighting or unclear images can make it difficult to identify the blood cells correctly.

3.3.3 Image Processing Module

The image processing modules use Python and OpenCV to analyse the captured blood sample images. The program loads the images and applies the image processing steps included in the code to obtain information about the blood cell counts.

The process may include converting the image into another format , improving its quality and examining areas where cells may be present . The information obtained is then used to calculate the NLR and MLR values.

The performance of this module depends on the image quality and the methods used to process it . Correctly identifying the neutrophils , lymphocytes and monocytes requires comparing the result with verified cell counts. Therefore the accuracy of the prototype must be tested before its results can be considered reliable.

3.3.4 Serial Communication Module

The serial communication module sends the processed cell count data from the computer to the Arduino UNO . After processing the blood sample image the Python program prepares the required values and send them through a serial connection .

The Arduino UNO receives these values and uses them for the NLR and MLR calculations. Proper communication settings and their consistent data format are important to ensure that the data are transferred correctly. This module allows the computer based image processing system to communicate with the Arduino.

3.3.5 Arduino Uno Processing Module

The Arduino UNO receives the blood count and calculates the NLR and MLR values based on the received data

NLR = Neutrophil Count ÷ Lymphocyte Count

MLR = Monocyte Count ÷ Lymphocyte Count

For example if the neutrophil count is 60 the lymphocyte count is 30 and the monocyte count is 6 the calculated NLR is 2.0 and the MLR is 0.2 . These numbers are only assembled and are not actual result obtained from the prototype

The Arduino UNO prepares the calculated values to be shown on the OLED display the program should also check the input data and prevent division by zero when calculating the ratios.

3.3.6 OLED Display Module

The display module presents the calculated NLR and MLR values using an OLED display controlled by the SH1106G.The Arduino Uno sends the calculated values to the display, allowing the output to be viewed locally. The display allows users to view the result easily. However the NLR and MLR values must be interpreted carefully as they cannot confirm the presence of cancer on their own.

3.4 Hardware Requirements

The hardware components help capture images of the blood sample, provide proper lighting, process the data and display the final results.

S. No.	Hardware Component	Purpose
1	OV7670 Camera	Captures images of the prepared sample
2	LED Illumination	Provides light during image acquisition
3	Arduino Uno	Calculates NLR and MLR and controls the display

4	SH1106G OLED Display	Displays the calculated ratios
5	Glass Slide	Holds the prepared sample for imaging
6	Computer	Runs the image acquisition and image-processing software
7	Sample Preparation Equipment	Supports sample handling according to the laboratory procedure

3.5 Software Requirements

The software components support image acquisition, image processing, serial communication and embedded programming.

S. No.	**Software**	**Purpose**
1	Python	Implements computer-based processing
2	OpenCV	Provides image-processing functions
3	Arduino IDE	Develops and uploads the Arduino program
4	Camera Acquisition Software	Captures images using the OV7670 camera setup
5	Serial Communication Interface	Transfers data between the computer and Arduino Uno

3.6 Overall Working Principle

The process start by preparing the blood sample and capturing its image using the OV7670 camera with LED lighting. The captured image is transferred to a computer, where Python and OpenCV process the image to obtain the required cell-count information. The processed data are sent to the Arduino UNO through serial communication. The Arduino uses the neutrophil ,lymphocyte and monocyte counts to calculate the NLR and MLR values the calculated results are shown on the SH1106G OLED display.

The proposed system brings together image capture , image processing on a computer and calculation using embedded hardware. It is designed to calculate and display NLR and MLR values and is not to diagnose cancer directly. Further the testing and comparison with standard laboratory results are needed to check the accuracy and reliability of the system.

CHAPTER 4

SYSTEM TESTING

4.1 Introduction to Testing

System testing is carried out to check whether the hardware and software components work properly and communicate with each other. The proposed system includes an OV7670 camera , LED lighting, Python and OpenCV, serial communication and an Ardunio Uno and an SH1106G OLED display. Each component list checked separately before testing the complete system.

The testing covers image capture , image processing, data transfer NLR and MLR calculation and displaying the results. It also checks whether the system handles invalid input correctly during the calculation.

The main purpose of testing is to find error checks that each module works as expected and ensure that the hardware and software work together properly. Further testing and comparison with standard laboratory results are needed to determine whether the cell counts and calculated ratios are accurate and reliable.

4.2 Module Testing

Model testing involves checking each part of the system separately to make sure it works correctly before connecting all the modules together.

4.2.1 Blood Sample Preparation Module Testing

The blood sample preparation module is tested to make sure the sample is ready for image capture. Proper preparation is important because the condition and visibility of the blood cells affect the quality of the image.

The prepared sample is placed on the glass slide using the required laboratory procedure. It is then viewed through the camera setup to check whether the cells are clearly visible for image processing.

During testing the samples checked for clear visibility, consistent preparation and unwanted marks or particles that may affect image analysis. Proper safety precautions must be followed when handling and disposing of blood samples.

4.2.2 Camera Module Testing

The camera modules is tested to check whether the OV7670 camera captures clear images of the prepared blood sample under LED lighting.

The camera is connected to the image capture setup and the software is used to take a picture of a sample . The captured image is then checked for clarity, brightness and overall quality.

This test helps find problems with poor lighting, incorrect camera placement or unsuitable camera settings. The captured image should be clear enough for the next stage of the image processing.

4.2.3 Image Processing Module Testing

The image processing module is tested using the image captured by the OV7670 camera. Python and OpenCV are used to open and process the image based on their method implemented in the program.

The image processing module is tested to check whether the program can open and process the image captured by the camera. During testing the output is checked for error, incorrect image changes and any problems that may occur during processing.

If the program provides white blood cell count they should be cumbered with manually verified counts or reliable reference data. This helps determine how accurate the system identifies neutrophils, lymphocytes and monocytes.

4.2.4 Serial Communication Module Testing

The serial communication module is tested to check whether data can be transferred correctly between the computer and the Arduino Uno.

The computer sends the required cell count through the serial connection. The Arduino UNO receives the data and checks whether the values are read in the correct format.

During testing the communication settings received values and the handling of incomplete are invalid data are checked. Proper data is transferred is important for calculating NLR and MLR correctly and displaying the results and the OLED screen.

4.2.5 Arduino Uno Module Testing

Arduino UNO module is tested to check whether it calculates the NLR and MLR correctly using the given blood counts.

The expected values are calculated manually and compared with the results produced by the Arduino program. Sample values can be used to check whether the calculation logic work properly.

For example if the neutrophil count is 60 the lymphocyte count is 30 and the monocyte count is 6 the expected calculations are

NLR=60/30=2.0

MLR=6/30=0.2

These values are only examples used to test the calculation process and were not obtained from an actual blood sample.

The program should also be tested with invalid inputs. If Lymphocyte count is 0 the system must prevent division by zero and should not display an incorrect ratio as a valid result.

4.2.6 OLED Display Module Testing

The OLED display modulus test to check whether the SH1106G display received and shows the values sent by the Arduino UNO.

The test check whether the display starts properly and show the NLR and MLR values clearly. It also verifies whether the displayed values match the result calculated by the Arduino.

The display should update whenever a new valid test is received. If the screen remains blank or shows incorrect value the connection display setting and programme should be checked.

4.3 Integration Testing

Integration testing is carried out to check whether all the modules work together properly as a complete system. It verifies the data pass correctly from image capture and processing to ratio calculation and final display.

The process starts with a blood sample preparation. The OV7670 camera capture an image of the prepared sample under LED lighting. The captured image is then transferred to a computer for processing.

Python and OpenCV process the image to obtain the required cell count. These values are sent to Arduino UNO through serial communication. The Arduino calculates the NLR and MLR values and display them on the OLED screen.

The integration test checks the following operations:

1. Successful capture of the prepared blood sample image.
2. Processing of the captured image by the computer.
3. Transfer of the required data through serial communication.
4. Correct calculation of NLR and MLR by the Arduino Uno.
5. Correct presentation of the calculated values on the OLED display.
6. Appropriate handling of invalid data and calculation errors.

The outcome of each test should be recorded during actual system operation. Any discrepancy between the expected and observed outputs should be investigated before the system is considered to be functioning correctly.

Table 4.1: System Testing

S.No.	Module Tested	Test Objective	Expected Result
1	Blood Sample Preparation	Check whether the prepared sample is suitable for imaging	Sample details are sufficiently visible for image acquisition
2	OV7670 Camera	Verify image acquisition under LED illumination	Sample image is captured
3	Python and OpenCV	Check image loading and processing	Image is processed without software errors
4	Serial Communication	Verify transfer of cell-count information	Arduino Uno receives the transmitted data correctly
5	Arduino Uno	Verify NLR and MLR calculations	Calculated ratios match the expected values for valid test inputs
6	OLED Display	Verify presentation of calculated ratios	NLR and MLR values are displayed correctly
7	Integrated System	Check the complete processing sequence	Data pass through the implemented modules and the output is displayed

CHAPTER 5

RESULT AND DISCUSSION

5.1 Introduction

This chapter presents the results and discussion of the proposed IoT-enabled system for cancer risk assessment and monitoring. The system combines blood sample preparation, image acquisition, image processing and embedded technology to support the calculation of two blood-based biomarkers: the Neutrophil-to-Lymphocyte Ratio (NLR) and the Monocyte-to-Lymphocyte Ratio (MLR).The prototype uses an OV7670 camera with LED illumination to capture images of the prepared blood sample. Python and OpenCV are used for image processing, while the Arduino Uno calculates NLR and MLR from the supplied cell-count information. The calculated values are presented on an OLED display controlled by the SH1106G.The results are discussed according to the operation of each module, the integration of the hardware and software components, and the limitations identified during development. The accuracy of the calculated NLR and MLR values depend on how correctly the system counts the blood cell from the image.

5.2 Blood Sample Preparation

Proper preparation of the blood sample is important because it affects the clarity of the captured image and the quality of the image processing. The sample is prepared according to the procedure used in the prototype and positioned on a glass slide for observation. When Turk's solution is used, it must be handled according to the approved laboratory procedure.

The prepared sample is placed under the camera arrangement with suitable LED illumination.



Figure 5.1: Blood Sample Preparation

The prepared blood samples used at the input for image capture preparing the sample consistently helps obtain clear images that can be used for further analysis.

5.3 Image Acquisition Results

The image acquisition module uses the OV7670 CMOS camera to capture the prepared blood sample. LED illumination provides light during image capture.

During the development process, the camera setup was used to capture a sample image. The captured image provides input for the Python-based image-processing stage.

Image quality depends on illumination, camera positioning, focus and sample preparation. Variations in these factors may affect the visibility of blood cells and the subsequent image analysis.

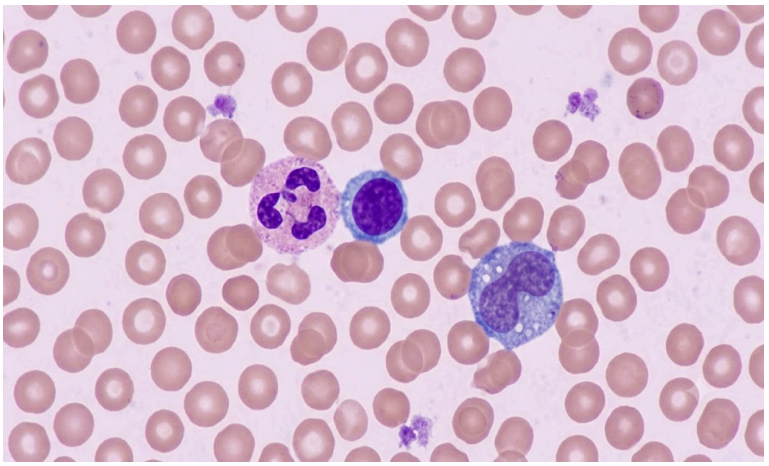


Figure 5.2: Blood Sample Image Captured Using OV7670 Camera

The captured image demonstrates the image acquisition stage of the prototype. However, successful image capture alone does not establish that individual white blood cell types can be identified accurately.

5.4 Image Processing Results

The captured image is processed using Python and the OpenCV library. The image-processing stage is intended to prepare the image for obtaining information about neutrophils, lymphocytes and monocytes.

Based on the processing method used, the software may load the captured image, and improve its quality and examine the areas where cells are present. The cell count information obtained from this process is then used to calculate NLR and MLR.

The results depend on the quality of the captured image and how well the image processing method works. To check the accuracy of the cell count they should be compared with manually counted cells or reliable reference data.

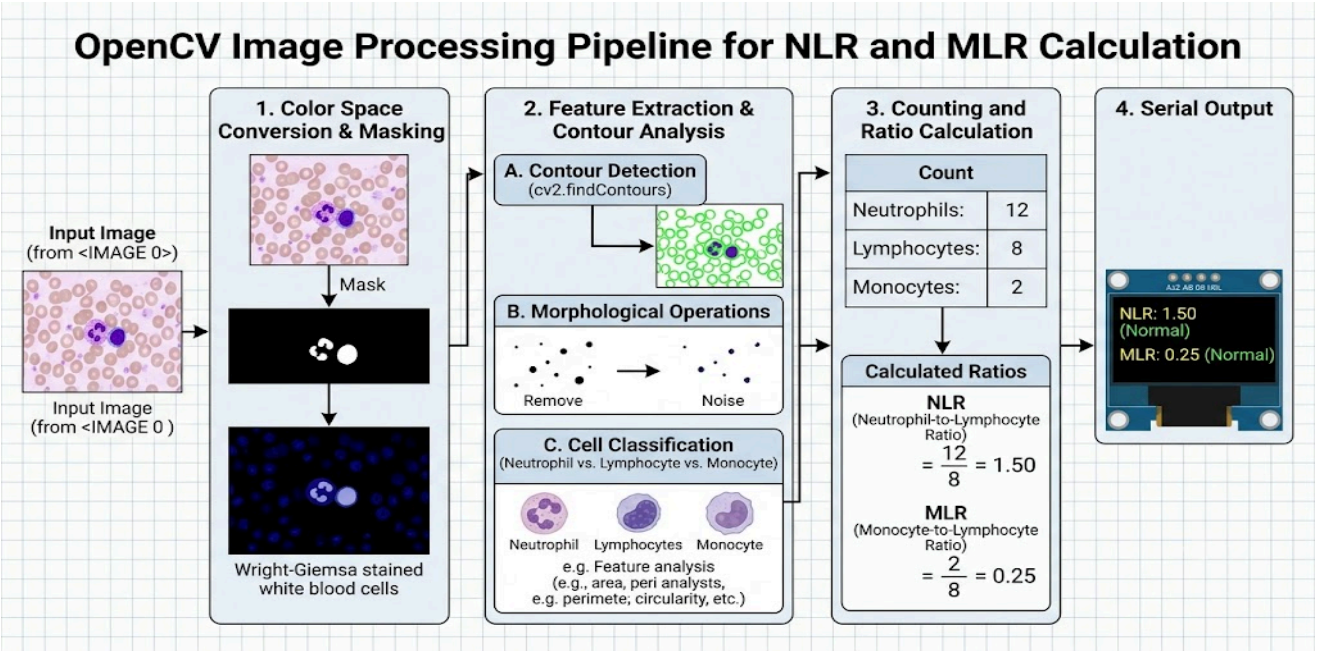


Figure 5.3: Image Processing Output Using Python and OpenCV

The image processing stage links image capture with the calculation process. If the system is not yet able to count the cells automatically and accurately, this part should be mentioned as still under development rather than presented as a fully automated cell classification system.

5.5 NLR and MLR Calculation Results

The Arduino UNO calculates the NLR and MLR values using the given neutrophil, lymphocyte and monocyte count.

The Neutrophil-to-Lymphocyte Ratio is calculated as: NLR=N/L

The Monocyte-to-Lymphocyte Ratio is calculated as: MLR=ML

Where:

- NN represents the neutrophil count.
- LL represents the lymphocyte count.
- MM represents the monocyte count.

For example, consider the following illustrative cell counts:

Cell Type	**Illustrative Count**
Neutrophils	60
Lymphocytes	30
Monocytes	6

The resulting ratios are:

NLR=60/30=2.00

MLR=6/30=0.20

These values are used only to explain how the calculation works and are not actual results obtained from the prototype.

Before calculating the ratios the program should check whether the received values are valid if the lymphocyte count is zero the program must prevent division by zero and should not display an incorrect result.

5.6 Hardware Integration and Output Display

The prototype integrates the image acquisition setup, computer-based image processing, serial communication, Arduino Uno and OLED display.

The computer processes the captured image and prepares the cell-count information. The information is transferred to the Arduino Uno through serial communication. The microcontroller calculates the ratios and sends the results to the SH1106G-controlled OLED display.

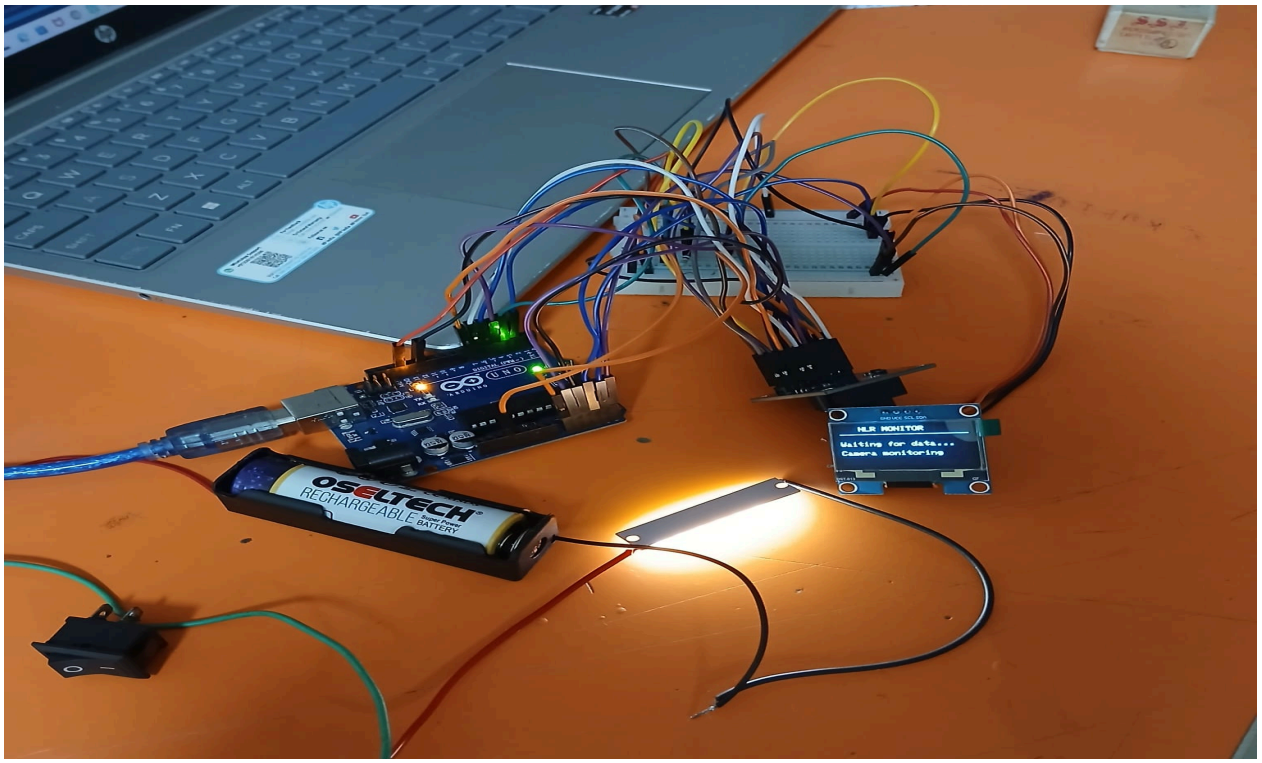


Figure 5.4: Arduino Uno and OLED Display Setup

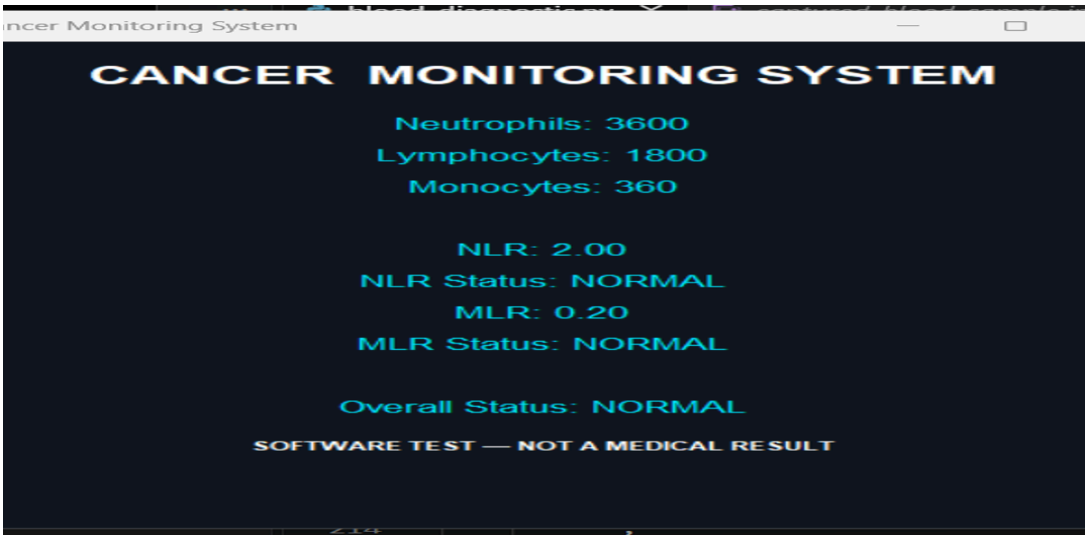


Figure 5.5: NLR and MLR Values Displayed on OLED

The OLED display provides a local method of viewing the calculated values. Correct display output depends on valid input data, successful serial communication, correct ratio calculation and proper display configuration.

The photographs and observations in this section should show how the actual prototype works.

5.7 System Testing Results

The system is tested by checking whether each module works properly on its own and whether all the modules work together as expected.

Table 5.1: Summary of System Testing

S. No.	Module	Test Objective	Result to Record
1	Blood Sample Preparation	Check sample suitability for imaging	Observed sample condition
2	OV7670 Camera	Verify sample image capture	Whether the image was captured successfully
3	Python and OpenCV	Check image processing	Actual processing output
4	Serial Communication	Verify transfer of count information	Whether the Arduino received valid data
5	Arduino Uno	Verify ratio calculation	Calculated values compared with expected values
6	OLED Display	Check result presentation	Whether the correct values appeared on the display
7	Integrated System	Evaluate the complete processing sequence	Actual outcome of the integration test

5.8 Discussion

The proposed system combines a camera , image processing software and electronic components to calculate NLR and MLR values .

The OV7670 camera captures the blood sample image which is then processed using Python and OpenCV. The processed data is sent to the Arduino Uno through serial communication. The Arduino calculates the NLR and MLR values and displays them on the OLED screen.

One of the main advantages of this system is that it brings these different functions together in a single prototype. However the accuracy of the result depends on obtaining correct blood cell count. Factor such as image clarity, lighting , sample preparation and proper identification of white blood cells can affect the results.

NLR and MLR values can also change due to health condition such as infection, inflammation. Therefore these values alone cannot confirm whether a person has cancer.

Further testing is required to cheque the performance of the prototype. The calculated cell counts and ratio should be compared with the reliable reference counts and standard laboratory result before considering any medical application.

5.9 Limitations

The current prototype has the following limitations:

1. **Image Quality:** Poor lighting, camera focus and improper sample preparation can affect the clarity of the captured images.
2. **Cell Identification:** Identifying neutrophils, lymphocytes and monocytes correctly depends on the image processing methods used and needs proper verification.
3. **Calculation Accuracy:** The NLR and MLR value depends on the accuracy of the blood cell count provided to the system.
4. **Clinical Interpretation:** NLR and MLR values can change due to infection inflammation and other health conditions therefore they cannot be used alone to confirm cancer.
5. **Validation:** More testing and comparison with standard laboratory results are needed to check the accuracy and reliability of the prototype.

5.10 Summary

This chapter covers the testing and the evaluation of the proposed system including the blood sample preparation, image capture, image processing NLR and MLR calculation and the connections of the hardware components.

The prototype brings together the OV7670 camera Python and OpenCV Arduino UNO and OLED display to calculate and show NLR and MLR values. However further testing and comparison to laboratory results are needed to check the accuracy of the calculation and determine whether the system can be used for healthcare monitoring.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 Conclusion

The project "**IoT -Enabled Multi-Biomarker Cancer Risk Assessment and Monitoring System**", aims to develop a prototype for assessing and monitoring cancer-related biomarkers using the Neutrophil-to-Lymphocyte Ratio (NLR) and Monocyte-to-Lymphocyte Ratio (MLR).

The system uses an OV7670 camera with LED lighting to capture blood sample images. Python and OpenCV are used to process the captured images and obtain the required cell-count information. These values are then sent to the Arduino UNO which calculates NLR and MLR. The calculated results are displayed in the OLED screen.

This project explores how image processing and electronic hardware can be combined for biomarker assessment. However further testing and comparison with standard laboratory results are needed to check the accuracy and reliability of the system. The prototype is intended to support health monitoring and is not designed to diagnose cancer independently.

6.2 Future Enhancement

The proposed system can be improved in the following ways:

1. Addition of Biomarkers: The system can be extended to calculate PLR (Platelet-to-Lymphocyte Ratio) and SII (Systemic Immune-Inflammation Index). Further research can help identify other biomarkers that may be useful for cancer risk assessment and monitoring.

2. Cloud Integration: Cloud Technology can be added to store biomarker values and patient records for long-term monitoring.

3.Doctor Dashboard: A secure dashboard can be developed to allow authorised doctors to view patient results remotely and track changes in biomarker values over time.

4.Improved Accuracy: The image-processing methods can be improved and the calculated results can be compared with standard laboratory measurements to check the accuracy of the system.

These improvements may make the system more useful and convenient for healthcare monitoring.

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